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IMAGE PROCESSING SYSTEM AND IMAGE PROCESSING METHOD

Abstract

An image processing system and an image processing method are provided. The image processing method includes: executing the following steps by an object detection module: receiving an image and detecting the image based on an object detection algorithm; and in response to detecting faces, obtaining a range of each face based on a shape of a polygon; executing the following steps by a weight calculation module on a plurality of pixels in a region covered by the ranges of all faces: based on coordinate information of a current pixel among the pixels and position information of each face, obtaining a weight of the current pixel; and in response to there being an unselected pixel among the pixels, selecting the unselected one of the pixels as the current pixel and returning to the aforementioned step of obtaining a weight of the current pixel.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This non-provisional application claims priority under 35 U.S.C. § 119 (a) to patent application No. 202410275803.1 filed in China, P.R.C. on Mar. 11, 2024, the entire contents of which are hereby incorporated by reference.

BACKGROUND

Technical Field

[0002] The present disclosure relates to the field of image processing, and in particular, to a technology for adjusting an image based on a face region.

Related Art

[0003] FIG. 1 is a schematic diagram of a face image. Referring to FIG. 1, when the HSI color gamut color and brightness of faces (a face **101** and a face **102**) in an image **100** are enhanced in the prior art, a transition region from the center of the face to surrounding parts is often uneven. Especially when a plurality of faces overlap, obvious color and brightness discontinuities are highly prone to appearance: when the image is played on TV, an obvious streak-shaped cut line (such as a cut line **103**) will appear at the edge of the face, and there will be obvious brightness and color discontinuities on both sides of the cut line.

SUMMARY

[0004] In view of this, some embodiments of the present disclosure provide an image processing system and an image processing method to improve the problems in the prior art.

[0005] An embodiment of the present disclosure provides an image processing system, including an object detection module and a weight calculation module, where the object detection module is configured to execute the following steps: (a1) receiving an image and detecting the image based on an object detection algorithm; and (a2) in response to detecting at least one face, obtaining a range of each face based on a shape of a polygon; where the weight calculation module is configured to execute the following steps on a plurality of pixels in a region covered by the ranges of all faces: (b1) based on coordinate information of a current pixel among the aforementioned pixels and position information of each face, obtaining a weight of the current pixel; and (b2) in response to there being an unselected pixel among the pixels, selecting the unselected one of the pixels as the current pixel and returning to step (b1).

[0006] An embodiment of the present disclosure provides an image processing method, including: (a) executing the following steps by an object detection module: (a1) receiving an image and detecting the image based on an object detection algorithm; and (a2) in response to detecting at least one face, obtaining a range of each face based on a shape of a polygon; (b) executing the following steps by a weight calculation module on a plurality of pixels in a region covered by the ranges of all faces: (b1) based on coordinate information of a current pixel among the aforementioned pixels and position information of each face, obtaining a weight of the current pixel; and (b2) in response to there being an unselected pixel among the pixels, selecting the unselected one of the pixels as the current pixel and returning to step (b1).

[0007] Based on the above, according to the image processing system and the image processing method provided by some embodiments of the present disclosure, the range of each face is obtained based on the shape of the polygon, and thereby the area of a non-face region and the area of a face overlapping region can be effectively reduced, and thus side effects brought by a non-face part

marked as a face region are alleviated; and the weight of a corresponding pixel obtained based on the coordinate information of each pixel in the region covered by the ranges of all faces and the position information of each face can be provided for the post-stage to adjust the brightness and chromaticity based on the weight of the corresponding pixel.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a schematic diagram of a face image;

[0009] FIG. 2 is a block diagram of an image processing system illustrated according to some embodiments of the present disclosure;

[0010] FIG. 3 is a schematic diagram of operation of an object detection module illustrated according to some embodiments of the present disclosure;

[0011] FIG. 4 is a schematic diagram of generation of face ranges illustrated according to some embodiments of the present disclosure;

[0012] FIG. 5A to FIG. 5D are schematic diagrams of operation of an image processing system illustrated according to some embodiments of the present disclosure;

[0013] FIG. 6 is a block diagram of a display system illustrated according to some embodiments of the present disclosure;

[0014] FIG. 7 is a schematic structural diagram of an electronic device illustrated according to some embodiments of the present disclosure;

[0015] FIG. 8 is a flowchart of an image processing method illustrated according to some embodiments of the present disclosure;

[0016] FIG. 9 is a flowchart of generation of face ranges illustrated according to some embodiments of the present disclosure;

[0017] FIG. 10 is a flowchart of an image processing method illustrated according to some embodiments of the present disclosure;

[0018] FIG. 11 is a flowchart of an image processing method illustrated according to some embodiments of the present disclosure;

[0019] FIG. 12 is a flowchart of an image processing method illustrated according to some embodiments of the present disclosure;

[0020] FIG. 13 is a flowchart of an image processing method illustrated according to some embodiments of the present disclosure;

[0021] FIG. 14 is a flowchart of an image processing method illustrated according to some embodiments of the present disclosure;

[0022] FIG. 15 is a flowchart of an image processing method illustrated according to some embodiments of the present disclosure; and

[0023] FIG. 16 is a flowchart of an image processing method illustrated according to some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0024] The aforementioned and other technical contents, features and effects of the present disclosure are clearly presented in the following detailed descriptions of embodiments with reference accompanying drawings. Anything that does not affect the efficacy that the present disclosure can produce and the purpose that can be achieved shall still fall within the scope of the technical content disclosed by the present disclosure.

[0025] FIG. 2 is a block diagram of an image processing system illustrated according to some embodiments of the present disclosure. FIG. 5A is a schematic diagram of operation of an image processing system illustrated according to some embodiments of the present disclosure. Referring to FIG. 2 and FIG. 5A, an image processing system **200** includes an object detection module **201**

and a weight calculation module **202**. The object detection module **201** is configured to receive an image **203** and detect the image **203** based on an object detection algorithm. The object detection algorithm is, for example, Faster R-CNN, SSD, YOLO or other object detection algorithms. The object detection module **201** is configured to obtain a range of each face based on a shape of a polygon when faces are detected in the image **203**.

[0026] Taking FIG. 5A as an example, in some embodiments, the polygon is an octagon. The object detection module **201** detects a face **101** and a face **102** in an image **300**. The object detection module **201**, in response to detecting the face **101** and the face **102**, obtains ranges of the face **101** and the face **102** based on a shape of the octagon, where the ranges of the face **101** and the face **102** are ranges included in an octagonal box **501** and an octagonal box **502**, respectively. The weight calculation module **202** is configured to calculate a weight for each pixel included in a region covered by ranges of all detected faces (e.g., the ranges of the face **101** and the face **102**).

[0027] The image processing method of some embodiments of the present disclosure and how modules of the image processing system **200** thereof cooperate with each other will be described below in details with the accompanying drawings.

[0028] FIG. 8 is a flowchart of an image processing method illustrated according to some embodiments of the present disclosure. Referring to FIG. 2, FIG. 5A and FIG. 8, in an embodiment of FIG. 8, the image processing method includes steps **S801** to **S808**. In step **S801**, an image (e.g., image **203**) is received by the object detection module **201** and the aforementioned image is detected based on an object detection algorithm to detect whether a face is present in the image or not. In step **S802**, if the object detection module **201** judges that no face is detected, then proceed to step **S803** to exit the current procedure to process a next image received by the object detection module **201**. If the object detection module **201** detects at least one face in the image, proceed to step **S804**. In step **S804**, in response to detecting at least one face, a range of each face is obtained based on a shape of a polygon (e.g., the octagon shown in FIG. 5A above) by the object detection module **201**. In step **S805**, for a current pixel which has not been unselected among pixels in a region covered by the ranges of all faces, based on coordinate information of the current pixel and position information of each face, a weight of the current pixel is obtained by the weight calculation module **202**.

[0029] In step **S806**, whether there is still an unselected pixel that is unselected for weight calculation among the pixels in the region covered by the ranges of all faces or not is judged by the weight calculation module **202**. If yes, proceed to step **S808**, and if no, proceed to step **S807** to exit the current procedure such that the remaining modules can process the pixels in the region covered by the ranges of all faces (e.g., brightness adjustment or chroma adjustment) by using weights of the pixels in the region covered by the ranges of all faces. In step **S808**, in response to there being an unselected pixel among the pixels in the region covered by the ranges of all faces, the unselected one of the pixels in the region covered by the ranges of all faces is selected as the current pixel by the weight calculation module **202** and returns to step **S805** to re-execute steps **S805** to **S808**.

[0030] FIG. 3 is a schematic diagram of operation of an object detection module illustrated according to some embodiments of the present disclosure. FIG. 4 is a schematic diagram of generation of face ranges illustrated according to some embodiments of the present disclosure. FIG. 9 is a flowchart of generation of face ranges illustrated according to some embodiments of the present disclosure. Referring to FIG. 2 to FIG. 4 and FIG. 9, in an embodiment of FIG. 9, the polygon is an octagon, and the image processing method includes executing steps **S901** to **S905** by the object detection module **201**. In step **S901**, a rectangular box of each face in a received image is obtained based on an object detection algorithm. Taking FIG. 3 as an example, the object detection module **201** detects the face **101** and the face **102** in the image **300**, and the object detection module **201** obtains a rectangular box **301** of the face **101** and a rectangular box **302** of the face **102** based on the object detection algorithm.

[0031] After step **S901**, rectangular boxes of detected faces are sequentially processed by the object

detection module **201** to obtain a range of each face. In step **S902**, based on four intercepts and four slopes corresponding to a current face among at least one detected face, four triangles at four corners of the rectangular box of the current face are deleted to obtain an octagonal box, and the object detection module **201** takes a range included in the octagonal box as the range of the current face. In step **S903**, whether there is an unselected face among the at least one detected face or not is judged. If no, proceed to step **S905** and exit the current procedure. If yes, execute step **904**. In step **904**, in response to there being an unselected face among the at least one face, the unselected one of the at least one face is selected as the current face and return to step **S902** to re-execute steps **S902** to **S905**.

[0032] Taking FIG. **4** as an example, the rectangular box of the current face obtained by the object detection module **201** is a rectangular box **400** (The rectangular box **400** may be the rectangular box **301** or the rectangular box **302**). The four intercepts corresponding to the current face are $d_{sub.1}$, $d_{sub.2}$, $d_{sub.3}$, and $d_{sub.4}$, respectively, and the four slopes are $m_{sub.1}$, $m_{sub.2}$, $m_{sub.3}$, and $m_{sub.4}$, respectively. The object detection module **201** can obtain a straight line **401** on the plane of the image **300** based on the intercept $d_{sub.1}$ and the slope $m_{sub.1}$. Similarly, the object detection module **201** can obtain straight lines **402** to **404** on the plane of the image **300** based on the intercepts $d_{sub.2}$, $d_{sub.3}$ and $d_{sub.4}$ and the corresponding slopes $m_{sub.2}$, $m_{sub.3}$, and $m_{sub.4}$. The object detection module **201** can obtain four triangles **4003** to **4006** at four corners of the rectangular box **400** of the current face based on intersections of the lines **401** to **404** and the rectangular box **400**. The object detection module **201** deletes the four triangles **4003** to **4006** at the four corners of the rectangular box **400** of the current face to obtain an octagonal box **405**. As shown in FIG. **5A**, after the rectangular box **301** and the rectangular box **302** are processed, an octagonal box **501** and an octagonal box **502** can be obtained.

[0033] In some embodiments of the present disclosure, the object detection module **201** obtains the intercepts $d_{sub.1}$, $d_{sub.2}$, $d_{sub.3}$, and $d_{sub.4}$ based on four ratios $r_{sub.1}$, $r_{sub.2}$, $r_{sub.3}$, and $r_{sub.4}$, and a length d of one side (upper side **4001**) of the rectangular box **400** of the current face, where $d_{sub.1}=r_{sub.1}d$, $d_{sub.2}=r_{sub.2}d$, $d_{sub.3}=r_{sub.3}d$, and $d_{sub.4}=r_{sub.4}d$.

[0034] In some embodiments of the present disclosure, the object detection module **201** multiplies lengths d (an upper side **4001** and a lower side **4002** have a same length) of upper and lower sides (upper side **4001** and lower side **4002**) of the rectangular box **400** by a preset ratio r to obtain the intercepts $d_{sub.1}$, $d_{sub.2}$, $d_{sub.3}$, and $d_{sub.4}$ (where $d_{sub.1}=d_{sub.2}=d_{sub.3}=d_{sub.4}=rd$). The object detection module **201** sets the slopes $m_{sub.1}$, $m_{sub.2}$, $m_{sub.3}$, and $m_{sub.4}$ with an absolute value of a preset slope. For example, the absolute value of the preset slope is m . Since the slope of the line **401** in a coordinate system illustrated in FIG. **4** is negative, the object detection module **201** sets $m_{sub.1}=-m$, and since the slope of the line **404** is positive, the object detection module **201** sets $m_{sub.4}=m$. Based on the same reason, the object detection module **201** sets $m_{sub.2}=m$, and $m_{sub.3}=-m$. In some embodiments of the present disclosure, the preset ratio r is 0.25, and the absolute value of the preset slope m is 0.25.

[0035] It is worth noting that based on the previous flows, boxes of other polygons can also be obtained, and the present disclosure does not limit a range of a face with an octagonal box. A user shall set an appropriate polygon according to usage requirements.

[0036] FIG. **10** is a flowchart of an image processing method illustrated according to some embodiments of the present disclosure. Referring to both FIG. **8** and FIG. **10**, in an embodiment of FIG. **10**, the coordinate information of the current pixel includes uv coordinate information, and the position information of each face includes uv coordinate information of a uv center and uv range information, where the uv center of the face is the center of the face in uv coordinates, and the uv range information includes a uv range value, representing a range of the face in a uv coordinate system. The aforementioned uv coordinate system is a texture coordinate system. Since the texture coordinate system is a two-dimensional coordinate system, each point in the texture coordinate system includes a first coordinate value and a second coordinate value. The uv coordinate

information included in the coordinate information of the current pixel is texture coordinate information of the current pixel. In this embodiment, when at least one face is detected in an image by the object detection module **201**, the object detection module **201** also detects the uv coordinate information of the uv center and the uv range information of each face based on an object detection algorithm.

[0037] Step **S805** includes executing step **S1001** by the weight calculation module **202**. In step **S1001**, based on the uv coordinate information of the uv center of each face and the uv coordinate information of the current pixel, for the uv center of each face, a uv distance from the uv center of the face to the current pixel is calculated to obtain a uv distance from the uv center of each face to the current pixel. Finally, based on the uv distances of all faces, a weight of the current pixel is obtained.

[0038] FIG. **11** is a flowchart of an image processing method illustrated according to some embodiments of the present disclosure. Referring to FIG. **8**, FIG. **10** to FIG. **11**, following an embodiment of FIG. **10**, when the object detection module **201** detects at least one face in an image, the object detection module **201** also detects the uv range information of each face based on an object detection algorithm and stores the uv range information in a memory for subsequent use. That is, in this embodiment, after the object detection module **201** detects the uv range information of each face based on the object detection algorithm, the uv range information will be stored in the memory for each detected face. In this embodiment, step **S1001** includes executing steps **S1101** to **S1107** by the weight calculation module **202**. In step **S1101**, based on the uv coordinate information of the uv center of a current face among faces and the uv coordinate information of the current pixel, the uv distance from the uv center of the current face to the current pixel is calculated.

[0039] In step **S1102**, whether the uv distance from the current pixel to the uv center of the current face is greater than a uv range value in the uv range information of the current face or not is judged. If yes, execute step **S1103**, and if no, execute step **S1104**. In step **S1103**, a ratio corresponding to the current face is set as a preset ratio, that is, the weight calculation module **202** sets the ratio corresponding to the current face with the preset ratio.

[0040] In step **S1104**, in response to the uv distance from the current pixel to the uv center of the current face being less than or equal to the uv range value in the uv range information, the ratio corresponding to the current face is set as: the preset ratio multiplied by the uv distance divided by the uv range value of the current face. In step **S1105**, whether there is an unselected face among the faces or not is judged. If yes, execute step **S1106**, and if no, execute step **S1107**. In step **S1106**, in response to there being an unselected face among the faces, the unselected one of the faces is selected as the current face and return to step **S1101** to further execute steps **S1101** to **S1107**. In step **S1107**, in response to the faces being all selected, a minimum one which is selected from the ratios corresponding to all faces is selected as the weight of the current pixel. In some embodiments of the present disclosure, the preset ratio is 64.

[0041] FIG. **12** is a flowchart of an image processing method illustrated according to some embodiments of the present disclosure. Referring to FIG. **8**, FIG. **10** to FIG. **12**, following an embodiment of FIG. **10**, when the object detection module **201** detects at least one face in an image, the object detection module **201** also detects the uv range information of each face based on an object detection algorithm and stores the uv range information in a memory for subsequent use. That is, in this embodiment, after the object detection module **201** detects the uv range information of each face based on the object detection algorithm, the uv range information will be stored in the memory for each detected face. In this embodiment, step **S1001** includes executing steps **S1201** to **S1206** by the weight calculation module **202**. In step **S1201**, based on the uv coordinate information of the uv center of a current face among faces and the uv coordinate information of the current pixel, the uv distance from the uv center of the current face to the current pixel is calculated.

[0042] In step **S1202**, whether the uv distance from the current pixel to the uv center of the current face is greater than a uv range value in the uv range information of the current face or not is judged. If yes, execute step **S1204**, and if no, execute step **S1203**. In step **S1203**, in response to the uv distance from the current pixel to the uv center of the current face being less than or equal to the uv range value in the uv range information of the current face, the ratio corresponding to the current face is set as: the preset ratio multiplied by the uv distance divided by the uv range value of the current face, and the ratio corresponding to the current face is put into a candidate weight set. That is, let $uv_{distance}$ represents the uv distance from the current pixel to the uv center of the current face, $uv_{range_currentface}$ represents the uv range value of the current face, and $preratio$ represents the preset ratio, then the ratio of the current face is set as

$$[00001] preratio \times \frac{uv_{distance}}{uv_{range_currentface}}.$$

[0043] In step **S1204**, whether there is an unselected face among the faces or not is judged. If yes, execute step **S1205**, and if no, execute step **S1206**. In step **S1205**, in response to there being an unselected face among the faces, the unselected one of the faces is selected as the current face and return to step **S1201** to further execute steps **S1201** to **S1206**. In step **S1206**, in response to the faces being all selected, all elements of the candidate weight set are averaged to obtain an average ratio, and the average ratio is set as the weight of the current pixel.

[0044] FIG. 13 is a flowchart of an image processing method illustrated according to some embodiments of the present disclosure. Referring to FIG. 11, FIG. 12, and FIG. 13, the uv coordinate information of the uv center of the current face includes a first coordinate value and a second coordinate value, and the uv coordinate information of the current pixel includes a first coordinate value and a second coordinate value. When the object detection module 201 detects at least one face in an image, the object detection module 201 also detects the first coordinate value and the second coordinate value of uv coordinates of the uv center of each face based on an object detection algorithm. The step of calculating the uv distance from the uv center of the current face to the current pixel in steps **S1101** and **S1201** include steps **S1301** to **S1305**.

[0045] In step **S1301**, an absolute value of a difference between the first coordinate value of the current pixel and the first coordinate value of the uv center of the current face is calculated to obtain a first absolute value. In step **S1302**, an absolute value of a difference between the second coordinate value of the current pixel and the second coordinate value of the uv center of the current face is calculated to obtain a second absolute value. In step **S1303**, whether the second absolute value is greater than the first absolute value or not is judged. If yes, execute step **S1304**, and if no, execute step **S1305**. In step **S1304**, in response to the second absolute value being greater than the first absolute value, the uv distance from the uv center of the current face to the current pixel is set as: the first absolute value multiplied by $\frac{1}{2}$ plus the second absolute value. In step **S1305**, in response to the second absolute value being less than or equal to the first absolute value, the uv distance from the uv center of the current face to the current pixel is set as: the second absolute value multiplied by $\frac{1}{2}$ plus the first absolute value.

[0046] If the first coordinate value of the current pixel is represented with $f_{sub.u}$, the second coordinate value of the current pixel is represented with $f_{sub.v}$, the first coordinate value of the uv center of the current face is represented with $center_{sub.u}$, the second coordinate value of the uv center of the current face is represented with $center_{sub.v}$, the absolute value of the difference between the first coordinate value of the current pixel and the first coordinate value of the uv center of the current face is represented with $t_{sub.u}$, the absolute value of the difference between the second coordinate value of the current pixel and the second coordinate value of the uv center of the current face is represented with $t_{sub.v}$, and the uv distance from the uv center of the current face to the current pixel is represented with $uv_{distance}$, then steps **S1301** to **S1305** can be represented with the following C++ pseudocode:

$$[00002] t_u = \text{abs}(f_u - center_u) \quad t_v = \text{abs}(f_v - center_v);$$

$uv_{distance} = t_v - t_u ? (t_v + t_u / 2) : (t_u + t_v / 2);$ [0047] where $\text{abs}()$ is a function provided by the

C++ language to calculate absolute values.

[0048] FIG. **14** is a flowchart of an image processing method illustrated according to some embodiments of the present disclosure. Referring to both FIG. **8** and FIG. **14**, the polygon is an octagon. The coordinate information of the current pixel includes pixel coordinate information. The position information of each face includes upper left corner coordinate information and lower right corner coordinate information of a rectangular box, four intercepts, four slopes, and region range information. The object detection module **201** obtains a rectangular box of each face in a received image, the upper left corner coordinate information and the lower right corner coordinate information of the rectangular box, and the region range information based on an object detection algorithm. The region range information includes a region range value, representing a range of a region covered by the faces. The pixel coordinate information included in the coordinate information of the current pixel includes pixel coordinates of the current pixel. The upper left corner coordinate information of the rectangular box of each face includes pixel coordinates of the upper left corner of the rectangular box, and the lower right corner coordinate information of the rectangular box of each face includes pixel coordinates of the lower right corner of the rectangular box, where the pixel coordinates are coordinates in pixels in the image. The pixel coordinates of each point in the received image include a first coordinate value and a second coordinate value. Four intercepts and slopes of each face are used for deleting four triangles at four corners of the rectangular box of the face to obtain an octagonal box. In this embodiment, the weight calculation module **202** stores a preset edge range for each face.

[0049] Step **S805** includes executing step **S1401** by the weight calculation module **202**. In step **S1401**, based on the upper left corner coordinate information and the lower right corner coordinate information of the rectangular box, the four intercepts, the four slopes and the region range information in the position information of each face, the edge range of each of the at least one face and the pixel coordinate information of the current pixel in the position information of each face, at least one ratio of the current pixel corresponding to at least one face is obtained by the weight calculation module **202**.

[0050] FIG. **5B** to FIG. **5C** are schematic diagrams of operation of an image processing system illustrated according to some embodiments of the present disclosure. Referring to FIG. **5B** to FIG. **5C** and FIG. **14**, following an embodiment of FIG. **14**, the object detection module **201** detects N faces in the image **203**, and the weight calculation module **202** executes the following steps: (1) storing the first coordinate value and the second coordinate value of the pixel coordinates of the upper left corner of the rectangular box of each face detected by the object detection module **201** in an array $starti[]$ and an array $startj[]$, respectively; (2) storing the first coordinate value and the second coordinate value of the pixel coordinates of the lower right corner of the rectangular box of each face detected by the object detection module **201** in an array $endi[]$ and an array $endj[]$, respectively; (3) storing the intercept and slope (e.g., $d.sub.1$ and $m.sub.1$ in FIG. **4**) of the upper left corner of a corresponding rectangular box of each face detected by the object detection module **201** in an array $jLU[]$ and an array $jLUm[]$, respectively, storing the intercept and slope (e.g., $d.sub.2$ and $m.sub.2$ in FIG. **4**) of the lower left corner of a corresponding rectangular box of each face detected by the object detection module **201** in an array $jLD[]$ and an array $jLDm[]$, respectively, storing the intercept and slope (e.g., $d.sub.4$ and $m.sub.4$ in FIG. **4**) of the upper right corner of a corresponding rectangular box of each face detected by the object detection module **201** in an array $jRU[]$ and an array $jLRm[]$, respectively, and storing the intercept and slope (e.g., $d.sub.3$ and $m.sub.3$ in FIG. **4**) of the lower right corner of a corresponding rectangular box of each face detected by the object detection module **201** in an array $jRD[]$ and an array $jRDm[]$, respectively; and (4) storing the region range value included in the region range information of each face detected by the object detection module **201** in an array $AI_region_range[]$. In this embodiment, the weight calculation module **202** stores an edge range for each face, and the aforementioned edge ranges corresponding to each face is a preset value and stored in the array

AI_edge_range[]. Since the aforementioned data is stored in arrays, a corresponding numerical value can be obtained by an index value of a corresponding face. For example, jLD[k] represents the intercept of the lower left corner of a corresponding rectangular box of the face with the index value of k, and AI_edge_range[k] represents the edge range of the face with the index value of k. In this embodiment, since the object detection module **201** detects the N faces in the image **203**, the range of k is from 0 to N-1.

[0051] In this embodiment, the weight calculation module **202** sets an array aiRatio6face[] to store the ratio of the current pixel corresponding to each face. The first coordinate value of the current pixel is represented with xi. The second coordinate value of the current pixel is represented with xj, and meanwhile, a variable zdist and finalRatio are set for calculation. Then the ratio of the current pixel corresponding to the face with the index value of k can be obtained with the following C++ pseudocode:

```
[00003] zdist = 0; z = starti[k] - xi, zdist = Max(zdist, z); z = xi - endi[k]; zdist = Max(zdist, z);  
z = startj[k] - xj, zdist = Max(zdist, z); z = xj - endj[k]; zdist = Max(zdist, z);  
z = startj[k] + jLU[k] - xj; z = starti[k] + (z / pow(2, 1 + jLUm[k])) - xi; zdist = Max(zdist, z);  
z = startj[k] + jRU[k] - xj; z = xi - endi[k] + (z / pow(2, 1 + jRUM[k])); zdist = Max(zdist, z);  
z = xj - endj[k] + jLD[k]; z = starti + (z / pow(2, 1 + jLDm[k])) - xi; zdist = Max(zdist, z);  
z = xj - endj[k] + jRD[k]; z = xi - endi[k] + (z / pow(2, 1 + jRDM[k])); zdist = Max(zdist, z);  
if zdist <= AI_edge_range[k]: if zdist == 0: finalRatio = 0; else:
```

```
finalRatio = 64 * (zdist / AI_edge_range[k]); else: finalRatio = 64; aiRatio6face[k] = finalRatio;
```

[0052] where Max() is a function provided by the C++ language, and its function is to take a parameter with a larger value from parameters and return the taken parameter. Pow() is a function provided by the C++ language, and its function is to return a result of an exponential operation taking a second parameter as an exponent of a first parameter. For example, pow(a,b) will return a.sup.b. By Letting k=0, 1, 2 . . . N-1 sequentially and executing the aforementioned pseudocode, the ratio of the current pixel corresponding to each face can be obtained. As can be seen from the aforementioned C++ pseudocode, if the current pixel is located within the octagonal box, zdist=0, and at this time, the ratio of the current pixel corresponding to the face is 0. If the current pixel is not located within an octagonal box, zdist is not 0. In a case where zdist is not 0, the ratio of the current pixel corresponding to the face will be set according to the ratio of zdist to

AI_edge_range[k]. The setting of finalRatio will be further explained with reference to FIG. 5B to FIG. 5D. A face **503** is a face detected by the object detection module **201**. The object detection

module **201** obtains an octagonal box **504** of the face **503** based on the aforementioned steps.

Referring to FIG. 5C, in an example of FIG. 5C, the edge range corresponding to the face **503** is set as 0. At this time, based on the aforementioned C++ pseudocode, it can be seen that the ratio of each pixel within a range **505** defined by the octagonal box **504** corresponding to the face **503** is calculated as 0 (because zdist is 0 in the octagonal box **504**), and the ratio of each pixel in the range **505** represented in black corresponding to the face **503** is calculated as 0 with the aforementioned C++ pseudocode. The ratio of each pixel outside the octagonal box **504** corresponding to the face **503** is calculated as 64 with the aforementioned C++ pseudocode.

[0053] Referring to FIG. 5D, in an example of FIG. 5D, the edge range corresponding to the face **503** is set as a non-zero positive number (e.g., 32). At this time, based on the aforementioned C++ pseudocode, it can be seen that the ratio of each pixel within the range **505** defined by the octagonal box **504** corresponding to the face **503** is calculated as 0 (because zdist is 0 in the octagonal box **504**). In a case where zdist is not 0 but less than or equal to the edge range, the ratio of the current pixel corresponding to the face will be set as a value less than 64 but greater than 0 according to $64 * (zdist / AI_edge_range[k])$. In FIG. 5D, a range **506** represents a range where finalRatio is not 64. Regarding the values of 0 to 64 represented in gray scale within the range **506**, the closer to white, the closer to 64 the value of finalRatio is, and the closer to black, the closer to 0 the value of finalRatio is.

[0054] FIG. 15 is a flowchart of an image processing method illustrated according to some embodiments of the present disclosure. Referring to FIG. 8, FIG. 14, and FIG. 15, following an embodiment of FIG. 14, in step S805, after step S1401, the weight calculation module 202 executes step S1501 to obtain the weight of the current pixel. In step S1501, a minimum ratio is selected from the at least one ratio of the current pixel corresponding to at least one face (stored in the array aiRatio6face[]) as the weight of the current pixel.

[0055] FIG. 16 is a flowchart of an image processing method illustrated according to some embodiments of the present disclosure. Referring to FIG. 8, FIG. 14, and FIG. 16, following an embodiment of FIG. 14, in step S805, after step S1401, the weight calculation module 202 executes step S1601 to obtain the weight of the current pixel. In step S1601, at least one ratio of the current pixel corresponding to the at least one face (stored in the array aiRatio6face[]) is averaged to obtain an average ratio. The weight calculation module 202 sets the average ratio as the weight of the current pixel.

[0056] FIG. 6 is a block diagram of a display system illustrated according to some embodiments of the present disclosure. In an embodiment of FIG. 6, a display system 600 includes a master chip 601, a high definition multimedia interface (HDMI) signal source 602, a USB signal source 603, and a display panel 604. The master chip 601 includes an image processing system 200 and a drive system 6011. The master chip 601 receives video data from the HDMI signal source 602 or the USB signal source 603. The image processing system 200 takes each frame of the video data as the image 203, and processes the image 203 based on the obtained weight in a case where a face is detected, and then, a processed image 203 is transmitted to the drive system 6011 to drive the display panel 604 for display.

[0057] FIG. 7 is a schematic structural diagram of an electronic device illustrated according to some embodiments of the present disclosure. In this embodiment, an architecture of the master chip 601 is an electronic device 700 as shown in FIG. 7. The electronic device 700 includes a processing unit 701, an internal memory 702, and a non-volatile memory 703. The internal memory 702, for example, is a random-access memory (RAM). The processing unit 701 is a processor. The internal memory 702 and the non-volatile memory 703 are configured to store a program, where the program may include a program code. The program code includes a computer operating instruction. The processing unit 701 reads a corresponding computer program from the non-volatile memory 703 into the internal memory 702 and then runs the computer program for implementing the object detection module 201 and the weight calculation module 202 at a logic level.

[0058] Based on the above, according to the image processing system and the image processing method provided by some embodiments of the present disclosure, the range of each face is obtained based on the shape of the polygon, and thereby the area of a non-face region and the area of a face overlapping region can be effectively reduced, and thus side effects brought by a non-face part marked as a face region are alleviated; and the weight of a corresponding pixel is obtained based on the coordinate information of each pixel in the region covered by the ranges of all faces and the position information of each face, and thereby, brightness and chroma can be adjusted based on the weight of the corresponding pixel.

Claims

1. An image processing system, comprising an object detection module and a weight calculation module, wherein the object detection module is configured to execute the following steps: (a1) receiving an image and detecting the image based on an object detection algorithm; and (a2) in response to detecting at least one face, obtaining a range of each of the at least one face based on a shape of a polygon; where the weight calculation module is configured to execute the following steps on a plurality of pixels in a region covered by the ranges of all of the at least one face: (b1) based on coordinate information of a current pixel among the pixels and position information of

each of the at least one face, obtaining a weight of the current pixel; and (b2) in response to there being an unselected pixel among the pixels, selecting the unselected one of the pixels as the current pixel and returning to step (b1).

2. The image processing system according to claim 1, wherein the polygon is an octagon, and step (a2) comprises the following steps to obtain the range of each of the at least one face: (a21) obtaining a rectangular box of each of the at least one face based on the object detection algorithm; (a22) based on four intercepts and four slopes of a current face among the at least one face, deleting four triangles at four corners of the rectangular box of the current face to obtain an octagonal box, wherein a range comprised in the octagonal box is taken as the range of the current face; and (a23) in response to there being an unselected face among the at least one face, selecting the unselected one of the at least one face as the current face and returning to step (a22).

3. The image processing system according to claim 2, wherein the object detection module is configured to obtain the intercepts based on four ratios and a length of one side of the rectangular box of the current face.

4. The image processing system according to claim 1, wherein the coordinate information of the current pixel comprises uv coordinate information, and the position information of each of the at least one face comprises uv coordinate information of a uv center and uv range information, and step (b1) comprises: (b11) based on the uv coordinate information of the uv center of each of the at least one face and the uv coordinate information of the current pixel, for the uv center of each of the at least one face, calculating a uv distance to the current pixel; and based on the uv distances of all of the at least one face and the uv range information, obtaining a weight of the current pixel.

5. The image processing system according to claim 4, wherein the uv range information comprises a uv range value, and step (b11) comprises: (b111) based on the uv coordinate information of the uv center of a current face among the at least one face and the uv coordinate information of the current pixel, calculating the uv distance from the uv center of the current face to the current pixel; (b112) in response to the uv distance from the current pixel to the uv center of the current face being greater than the uv range value of the current face, setting a ratio corresponding to the current face as a preset ratio; and in response to the uv distance from the current pixel to the uv center of the current face being less than or equal to the uv range value of the current face, setting the ratio corresponding to the current face as: a preset ratio multiplied by the uv distance divided by the uv range value of the current face; and (b113) in response to there being an unselected face among the at least one face, selecting the unselected one of the at least one face as the current face and returning to step (b111); and in response to the at least one face being selected, selecting a minimum one from the ratios of all of the at least one face as the weight of the current pixel.

6. The image processing system according to claim 4, wherein the uv range information comprises a uv range value, and step (b11) comprises: (b111) based on the uv coordinate information of the uv center of a current face among the at least one face and the uv coordinate information of the current pixel, calculating the uv distance from the uv center of the current face to the current pixel; (b112) in response to the uv distance from the current pixel to the uv center of the current face being greater than the uv range value of the current face, executing step (b113); in response to the uv distance from the current pixel to the uv center of the current face being less than or equal to the uv range value of the current face, setting a ratio corresponding to the current face as: a preset ratio multiplied by the uv distance divided by the uv range value of the current face; and putting the ratio corresponding to the current face into a candidate weight set; and (b113) in response to there being an unselected face among the at least one face, selecting the unselected one of the at least one face as the current face and returning to step (b111); and in response to the at least one face being selected, averaging all elements in the candidate weight set to obtain an average ratio, and setting the average ratio as the weight of the current pixel.

7. The image processing system according to claim 5, wherein the uv coordinate information of the uv center of the current face comprises a first coordinate value and a second coordinate value, the

uv coordinate information of the current pixel comprises a first coordinate value and a second coordinate value, and the aforementioned step of calculating the uv distance from the uv center of the current face to the current pixel comprises: calculating an absolute value of a difference between the first coordinate value of the current pixel and the first coordinate value of the uv center of the current face to obtain a first absolute value; calculating an absolute value of a difference between the second coordinate value of the current pixel and the second coordinate value of the uv center of the current face to obtain a second absolute value; and in response to the second absolute value being greater than the first absolute value, setting the uv distance from the uv center of the current face to the current pixel as: the first absolute value multiplied by $\frac{1}{2}$ plus the second absolute value; and in response to the second absolute value being less than or equal to the first absolute value, setting the uv distance from the uv center of the current face to the current pixel as: the second absolute value multiplied by $\frac{1}{2}$ plus the first absolute value.

8. The image processing system according to claim 1, wherein the polygon is an octagon, corresponding to each of the at least one face, an edge range is preset, the coordinate information of the current pixel comprises pixel coordinate information, the position information of each of the at least one face comprises upper left corner coordinate information and lower right corner coordinate information of a rectangular box, four intercepts, four slopes, and region range information, and step (b1) comprises: (b11) based on the upper left corner coordinate information and the lower right corner coordinate information of the rectangular box, the four intercepts, the four slopes and the region range information in the position information of each of the at least one face, the edge range of each of the at least one face, and the pixel coordinate information of the current pixel, obtaining at least one ratio of the current pixel corresponding to the at least one face.

9. The image processing system according to claim 8, wherein step (b1) comprises: (b12) selecting a minimum ratio from the at least one ratio of the current pixel corresponding to the at least one face as the weight of the current pixel.

10. The image processing system according to claim 8, wherein step (b1) comprises: (b12) averaging the at least one ratio of the current pixel corresponding to the at least one face to obtain an average ratio, and setting the average ratio as the weight of the current pixel.

11. An image processing method, comprising: (a) executing, by an object detection module, the following steps: (a1) receiving an image and detecting the image based on an object detection algorithm; and (a2) in response to detecting at least one face, obtaining a range of each of the at least one face based on a shape of a polygon; and (b) executing, by a weight calculation module, the following steps on a plurality of pixels in a region covered by the ranges of all of the at least one face: (b1) based on coordinate information of a current pixel among the pixels and position information of each of the at least one face, obtaining a weight of the current pixel; and (b2) in response to there being an unselected pixel among the pixels, selecting the unselected one of the pixels as the current pixel and returning to step (b1).

12. The image processing method according to claim 11, wherein the polygon is an octagon, and step (a2) comprises the following steps to obtain the range of each of the at least one face: (a21) obtaining a rectangular box of each of the at least one face based on the object detection algorithm; (a22) based on four intercepts and four slopes of a current face among the at least one face, deleting four triangles at four corners of the rectangular box of the current face to obtain an octagonal box, wherein a range comprised in the octagonal box is taken as the range of the current face; and (a23) in response to there being an unselected face among the at least one face, selecting the unselected one of the at least one face as the current face and returning to step (a22).

13. The image processing method according to claim 12, wherein step (a2) comprises: obtaining, by the object detection module, the intercepts based on four ratios and a length of one side of the rectangular box of the current face.

14. The image processing method according to claim 11, wherein the coordinate information of the current pixel comprises uv coordinate information, and the position information of each of the at

least one face comprises uv coordinate information of a uv center and uv range information, and step (b1) comprises: (b11) based on the uv coordinate information of the uv center of each of the at least one face and the uv coordinate information of the current pixel, for the uv center of each of the at least one face, calculating a uv distance to the current pixel; and based on the uv distances of all of the at least one face and the uv range information, obtaining a weight of the current pixel.

15. The image processing method according to claim 14, wherein the uv range information comprises a uv range value, and step (b11) comprises: (b111) based on the uv coordinate information of the uv center of a current face among the at least one face and the uv coordinate information of the current pixel, calculating the uv distance from the uv center of the current face to the current pixel; (b112) in response to the uv distance from the current pixel to the uv center of the current face being greater than the uv range value of the current face, setting a ratio corresponding to the current face as a preset ratio; and in response to the uv distance from the current pixel to the uv center of the current face being less than or equal to the uv range value of the current face, setting the ratio corresponding to the current face as: a preset ratio multiplied by the uv distance divided by the uv range value of the current face; and (b113) in response to there being an unselected face among the at least one face, selecting the unselected one of the at least one face as the current face and returning to step (b111); and in response to the at least one face being selected, selecting a minimum one from the ratios of all of the at least one face as the weight of the current pixel.

16. The image processing method according to claim 14, wherein the uv range information comprises a uv range value, and step (b11) comprises: (b111) based on the uv coordinate information of the uv center of a current face among the at least one face and the uv coordinate information of the current pixel, calculating the uv distance from the uv center of the current face to the current pixel; (b112) in response to the uv distance from the current pixel to the uv center of the current face being greater than a uv range value of the current face, executing step (b113); and in response to the uv distance from the current pixel to the uv center of the current face being less than or equal to the uv range value of the current face, setting a ratio corresponding to the current face as: a preset ratio multiplied by the uv distance divided by the uv range value of the current face, and putting the ratio corresponding to the current face into a candidate weight set; and (b113) in response to there being an unselected face among the at least one face, selecting the unselected one of the at least one face as the current face and returning to step (b111); and in response to the at least one face being selected, averaging all elements in the candidate weight set to obtain an average ratio, and setting the average ratio as the weight of the current pixel.

17. The image processing method according to claim 15, wherein the uv coordinate information of the uv center of the current face comprises a first coordinate value and a second coordinate value, the uv coordinate information of the current pixel comprises a first coordinate value and a second coordinate value, and the aforementioned step of calculating the uv distance from the uv center of the current face to the current pixel comprises: calculating an absolute value of a difference between the first coordinate value of the current pixel and the first coordinate value of the uv center of the current face to obtain a first absolute value; calculating an absolute value of a difference between the second coordinate value of the current pixel and the second coordinate value of the uv center of the current face to obtain a second absolute value; and in response to the second absolute value being greater than the first absolute value, setting the uv distance from the uv center of the current face to the current pixel as: the first absolute value multiplied by $\frac{1}{2}$ plus the second absolute value; and in response to the second absolute value being less than or equal to the first absolute value, setting the uv distance from the uv center of the current face to the current pixel as: the second absolute value multiplied by $\frac{1}{2}$ plus the first absolute value.

18. The image processing method according to claim 11, wherein the polygon is an octagon, corresponding to each of the at least one face, an edge range is preset, the coordinate information of the current pixel comprises pixel coordinate information, the position information of each of the at

least one face comprises upper left corner coordinate information and lower right corner coordinate information of a rectangular box, four intercepts, four slopes, and region range information, and step (b1) comprises: (b11) based on the upper left corner coordinate information and the lower right corner coordinate information of the rectangular box, the four intercepts, the four slopes and the region range information in the position information of each of the at least one face, the edge range of each of the at least one face, and the pixel coordinate information of the current pixel, obtaining at least one ratio of the current pixel corresponding to the at least one face.

19. The image processing method according to claim 18, wherein step (b1) comprises: (b12) selecting a minimum ratio from the at least one ratio of the current pixel corresponding to the at least one face as the weight of the current pixel.

20. The image processing method according to claim 18, wherein step (b1) comprises: (b12) averaging the at least one ratio of the current pixel corresponding to the at least one face to obtain an average ratio, and setting the average ratio as the weight of the current pixel.
